MAS365 - Nijat Abbasov . 20230929.

Pr. 8.3

if $f(x)$ is a polynomial of degree less then n,

Qudrature rull ~~that~~ is based on exact interpolation ~~and~~ what I mean is $\varphi(f)$ will be exactly equal to $\int_0^1 f(x)dx$. so, I choose. $f(x)=1$ and write.

$$\int_0^1 dx \cdot 1 = \varphi(f) = \sum_{i=1}^{n} w_i f(x) = \sum_{i=1}^{n} w_i$$

$$1 = \sum w_i$$

Pr. 8.6

$\cancel{\varphi(f) = \sum w_i}$

$\cancel{f(x)}$ $p(x) = \sum_{i=1}^{n} f(x_i) \cdot l_i(x)$

$$\int_a^b p(x)dx = \int_a^b \sum_{i=1}^{n} f(x_i) \cdot l_i(x)dx = \sum_{i=1}^{n} \cdot \int_a^b f(x_i)\, l_i(x)dx =$$

$$= \sum_{i=1}^{n} f(x_i) \int_a^b l_i(x)dx =$$

$$= \sum_{i=1}^{n} w_i f(x_i)$$

$$w_i = \int_a^b l_i(x)dx$$

Pr. 8.5.

a) if $f''(x) \geq 0$ on $[a,b)$

that means the function is convex, so, proving b) means, a) is also true.

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Pr. 8.5.

b)

$$M_k(f) = \sum_{i=1}^{k} f\left(\frac{x_{i-1}+x_i}{2}\right)\Delta x_i.$$

$\Delta x_i = x_i - x_{i-1}$

~~if~~

$a < x_i \leq x_{i+1} < b$

$$T_k(f) = \sum_{i=1}^{k} \frac{f(x_{i-1})+f(x_i)}{2}\Delta x_i$$

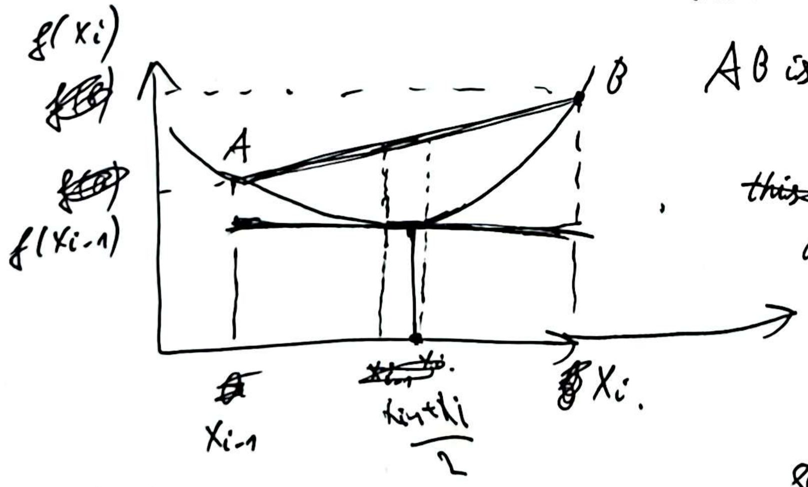
we can represent $\int_a^b f(x)dx$ as $\to \sum_{i=1}^{k}\int_{x_{i-1}}^{x_i} f(x)dx$,

prove:

$$\sum_{i=1}^{k} f\left(\frac{x_{i-1}+x_i}{2}\right)\Delta x_i \leq \sum_{i=1}^{k}\int_{x_{i-1}}^{x_i} f(x)dx \leq \sum_{i=1}^{k}\frac{f(x_{i-1})+f(x_i)}{2}\Delta x_i.$$

above holds if:

$$f\left(\frac{x_{i-1}+x_i}{2}\right)\Delta x_i \leq \int_{x_{i-1}}^{x_i} f(x)dx \leq \frac{f(x_{i-1})+f(x_i)}{2}\Delta x_i.$$ holds.



the area under line AB is $\frac{f(x_{i-1})+f(x_i)}{2}\Delta x_i$.

~~this area is~~ all the points on the line AB is above the function because of geometric meaning of convex function.

so, $\frac{f(x_{i-1})+f(x_i)}{2}\Delta x_i \geq \int_{x_{i-1}}^{x_i} f(x)dx$.

midpoint area is same as area induced by tangent line drawn at $\frac{x_{i-1}+x_i}{2}$ which is less than $\int_{x_{i-1}}^{x_i} f(x)dx$ due to convexity, so, $f\left(\frac{x_{i-1}+x_i}{2}\right)\Delta x_i \leq \int_{x_{i-1}}^{x_i} f(x)dx$.

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Pr. 9.8.

a) $y' = -y^2$. $y_0 = 1$ $h = 0.1$.

$y_1 = y_0 + (-y_1^2) \cdot h$

$y_1 = 1 - 0.1 y_1^2$

b) $y_1 = 1 - 0.1 y_1^2$.

$0.1 y_1^2 + y_1 - 1 = 0$

c) $f(y_1) = 0$

$$y_{1,i+1} = y_{1,i} - \frac{f(y_{1,i})}{f'(y_{1,i})} =$$

$y_{1s} \to y_1$ starting value.

$$= y_{1,i} - \frac{0.1 y_{1,i}^2 + y_{1,i} - 1}{0.2 y_{1,i} + 1},$$

$$y_{1s} = y_0 + h f'(y_0) =$$
$$= y_0 + h(-y_0^2) =$$
$$= y_0 - y_0^2 h = 1 - 0.1 = 0.9 = y_{1s}.$$

d) $$y_1 = y_{1s} - \frac{f(y_{1s})}{f'(y_{1s})} = y_{1s} - \frac{0.1 y_{1s}^2 + y_{1s} - 1}{0.2 y_{1s} + 1} =$$

$$= 0.9 - \frac{0.081 + 0.9 - 1}{1.18} =$$

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Pr. 9. 8.

d) $0.9 - \frac{0.081 - 0.1}{1.18} = 0.9 + \frac{0.019}{1.18} = 0.9161016949 \approx 0.92$

Pr. 9.1.

a) $y' = u$

$u' = t + y + u$

b) $y' = u$

$u' = w$

$w' = w + ty$

c) $y' = u$

$u' = w$

$w' = w - 2u + y - t + 1.$

Pr. 9.2.

a) $y' = u$

$u' = u(1 - y^2) - y$

b) $y' = u$

$u' = w$

$w' = -yw.$

c) $y_1' = u$

$y_2' = w.$

$u' = -GM \frac{y_1}{(y_1^2 + y_2^2)^{3/2}}.$

$w' = -GM \frac{y_2}{(y_1^2 + y_2^2)^{3/2}}$